

The background of the slide is a close-up, artistic rendering of several red blood cells. The cells are depicted in a deep red color with a textured, bumpy surface. They are arranged in a way that creates a sense of depth, with some cells in sharp focus in the foreground and others blurred in the background. The lighting highlights the three-dimensional shape of the cells, which are generally disc-shaped with a slight indentation in the center.

Heart Disease:

The Path to its Prediction

Exploring Use of Machine Learning
in Healthcare Data

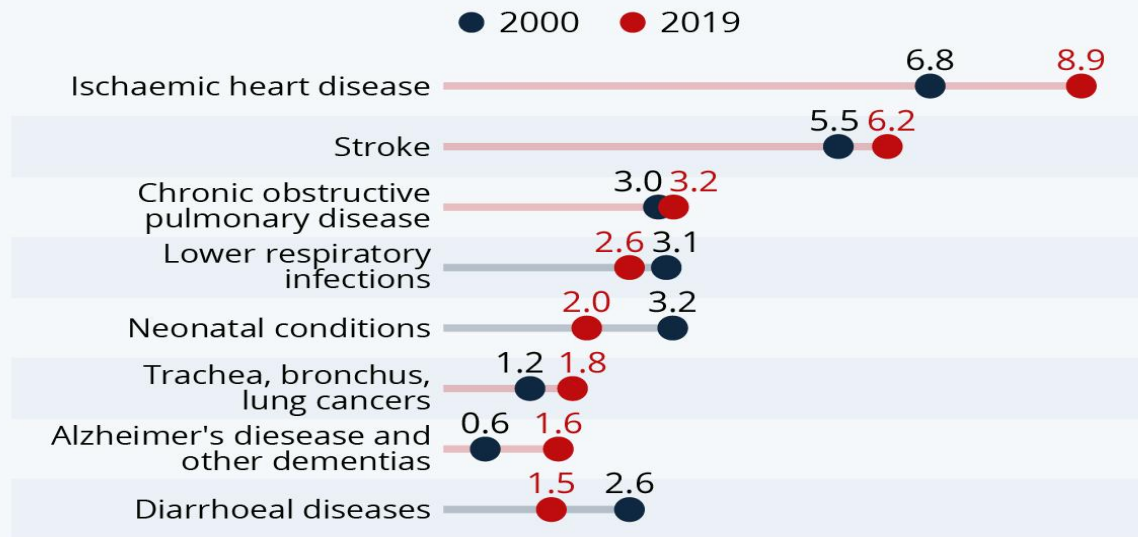
John Katsoulakis

January 19, 2026

Heart Disease and Stroke (HD) the leading causes of death globally

The World's Leading Causes Of Death

Total number of people who died from the following conditions (in millions)



Source: World Health Organization



statista

Why is prediction of a heart disease important ?

- The leading cause of death globally
- 9 million people died from heart disease in 2019
- 30% increase in the number of deaths over 20 years

What is this project about ?

- A student's attempt to predict accurately whether a person has a heart disease or not applying machine-learning modeling into real-world data

Ten different predictive variables of various types used

Both demographics (e.g., age) and health-related (e.g., total cholesterol)

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   918 non-null   int64
1   Sex                   918 non-null   string
2   ChestPainType         918 non-null   string
3   RestingBP             918 non-null   int64
4   Cholesterol            918 non-null   int64
5   FastingBS             918 non-null   int64
6   RestingECG            918 non-null   string
7   MaxHR                 918 non-null   int64
8   ExerciseAngina        918 non-null   string
9   Oldpeak               918 non-null   float64
10  ST_Slope              918 non-null   string

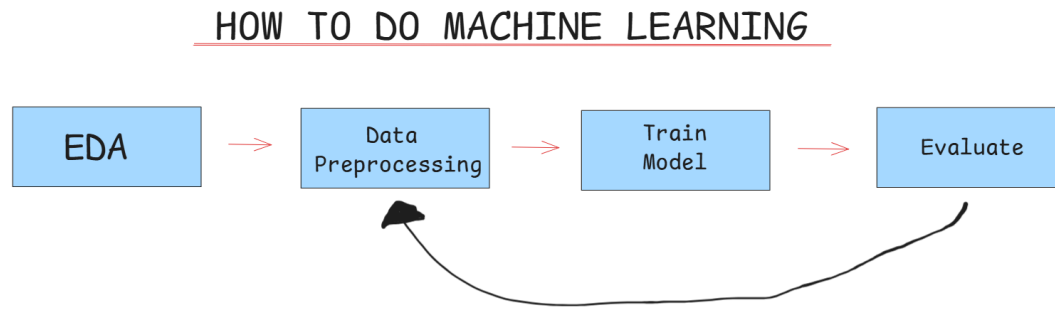
dtypes: float64(1), int64(6), string(5)
memory usage: 86.2 KB
```

Demographics

Health metrics

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	0
1	49	F	NAP	160	180	0	Normal	156	N	1.0	Flat	1
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	0
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	0

A four-step data process leveraging Machine Learning



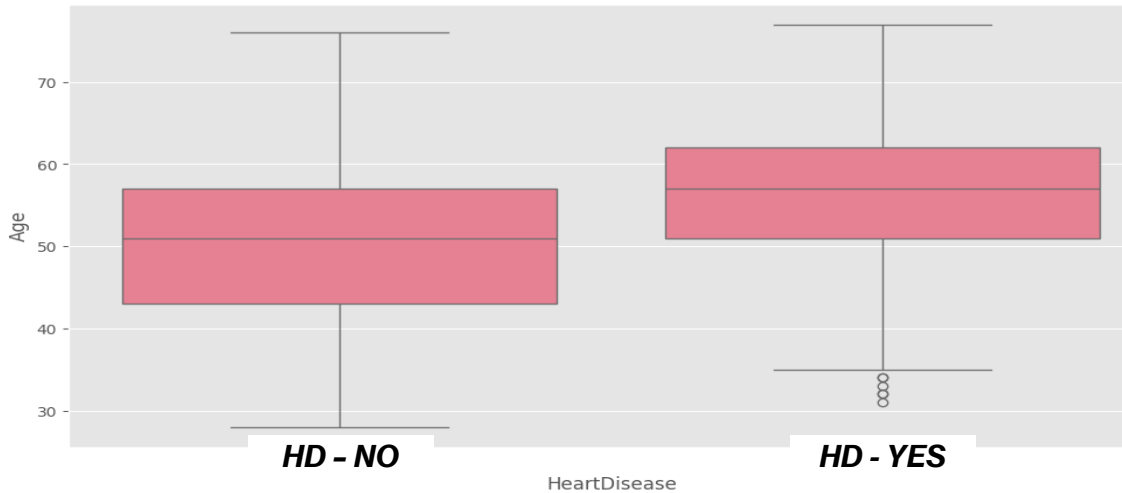
A four-step process

1. Exploratory Data Analysis (EDA)
2. Data Cleaning & Preprocessing
3. Model Selection & Training
4. Results & Evaluation

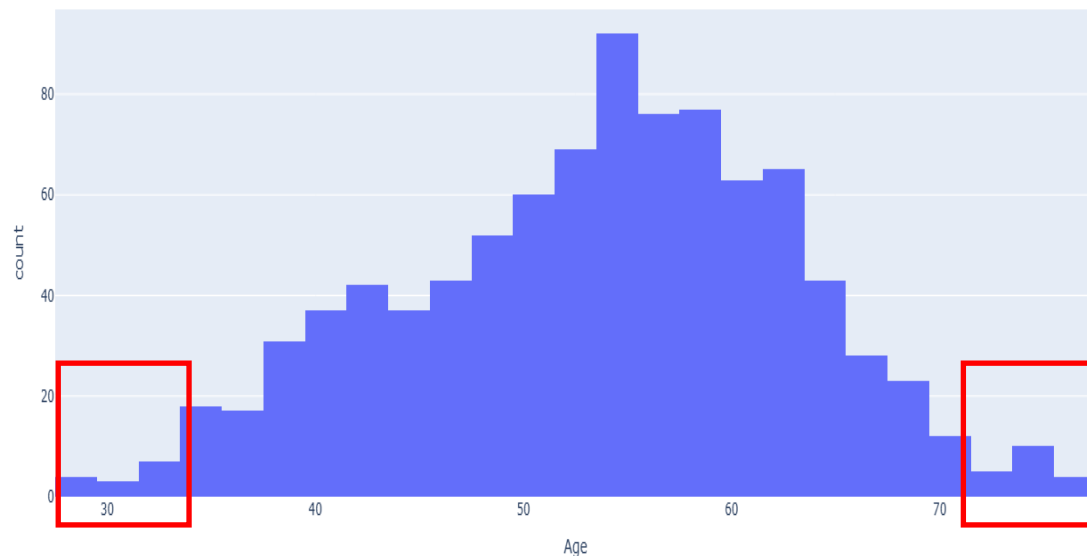
EDA: Analyzing the data distribution for each variable and...

...checking for outliers

Data Analysis : Age distribution by people-group



Data Analysis : Age outliers in patients with heart disease

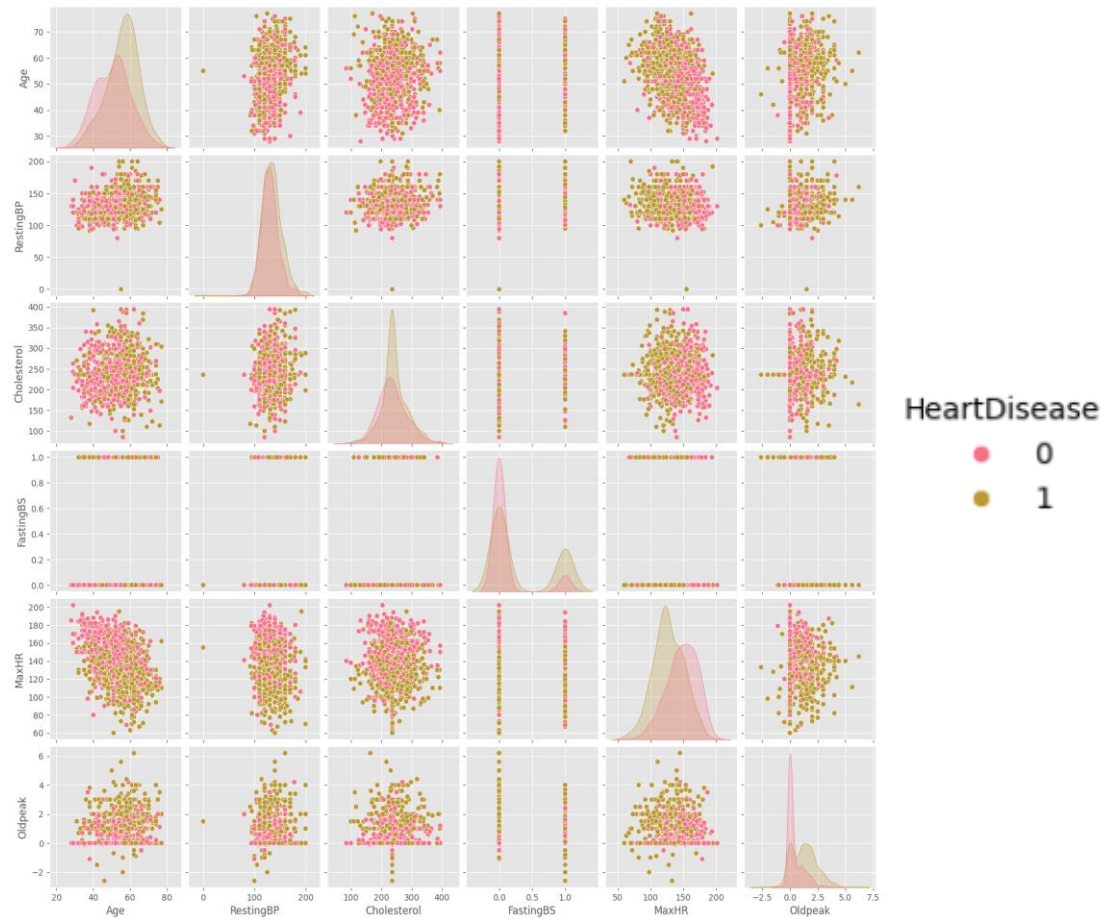


- 57 y.o. the median age for patients with heart disease
- Those with heart disease are, on average, older by 5 years
- Still there are four outliers – patients younger than 36 years old having heart disease
- Majority of patients with heart disease between the ages of 52 and 62
- Still, there are four outliers – patients younger than 35 years old having heart disease

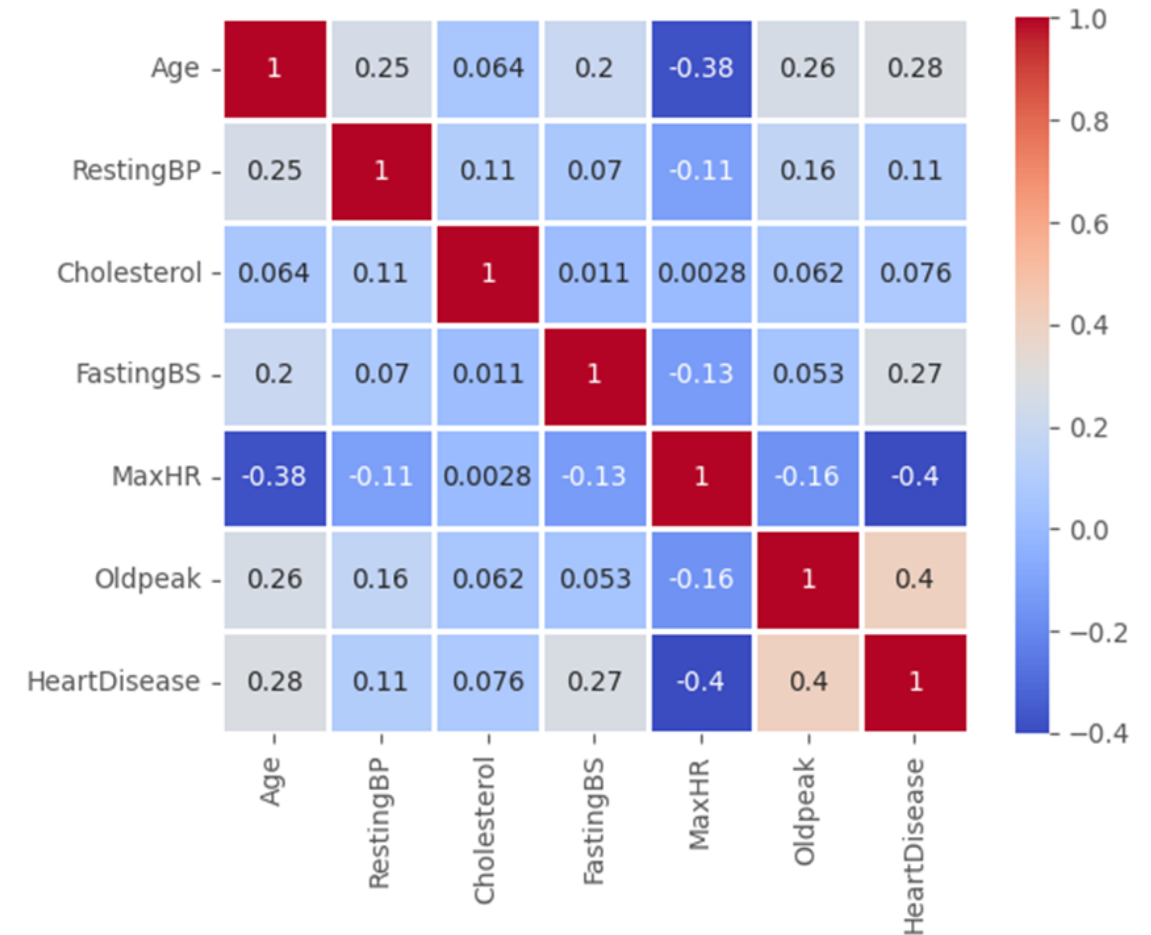
EDA: Using pair-plots and correlation tables

Oldpeak the best single predictor of Heart Disease followed by Age

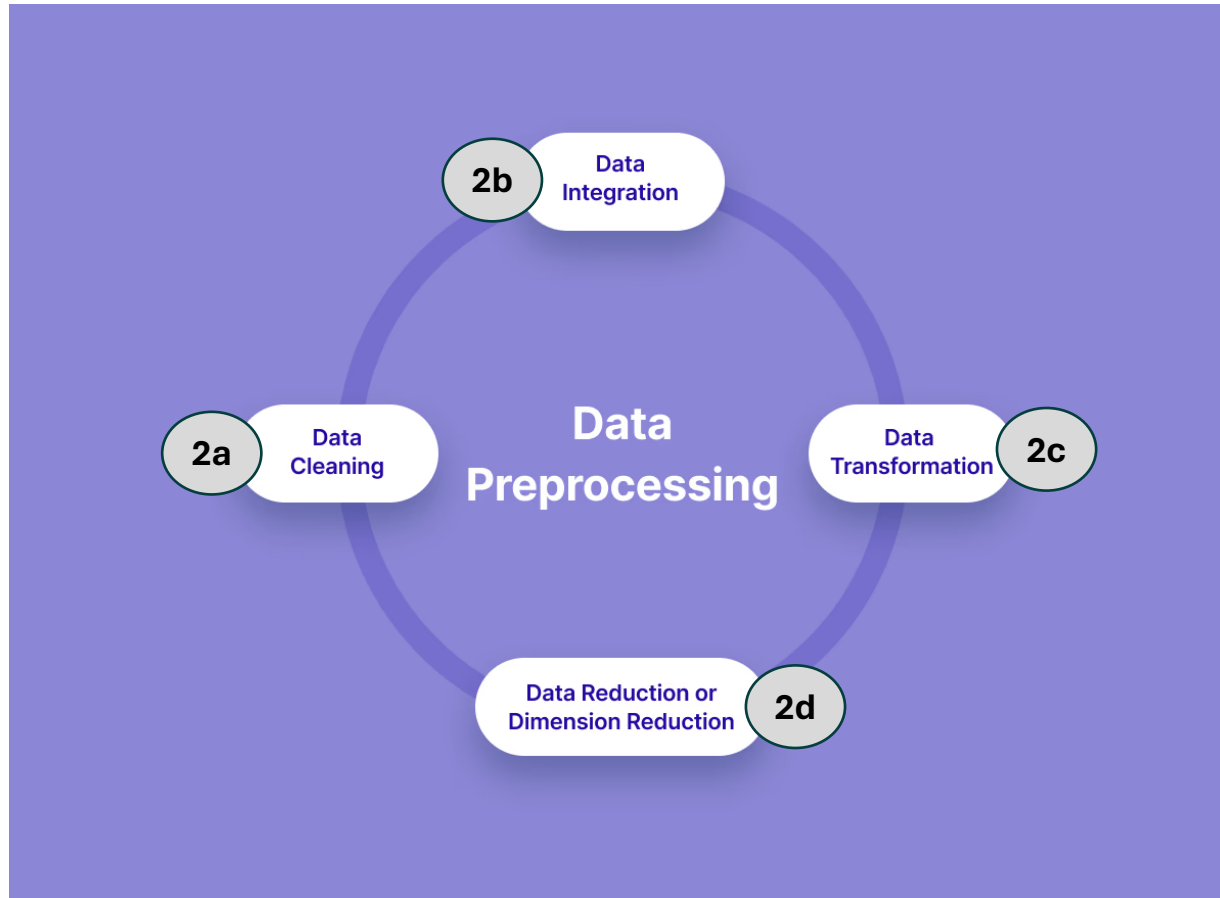
Pair-Plots



Correlation tables



Data cleaning and pre-processing: 4 sub-steps followed



Data...

- Cleaned for any anomalies or wrong texting
- Integrated into the database replacing zeros or null values with means or averages
- Transformed through scaling (e.g., logarithmic transformation) to smoothen out differences
- Reduced if, even after scaling, unusual outliers exist that may generate bias to the model

Model Selection and Training

6 ML models used to predict whether a person has heart disease or not

3a

Tree-Based Models

- **Decision Trees**
- **Random Forests**

3b

Non-Tree Based Models

- **K Nearest Neighbors (KNN)**
- **Support Vector Machine (SVM)**
- **Logistic Regression**
- **Naïve Bayes Classifier**

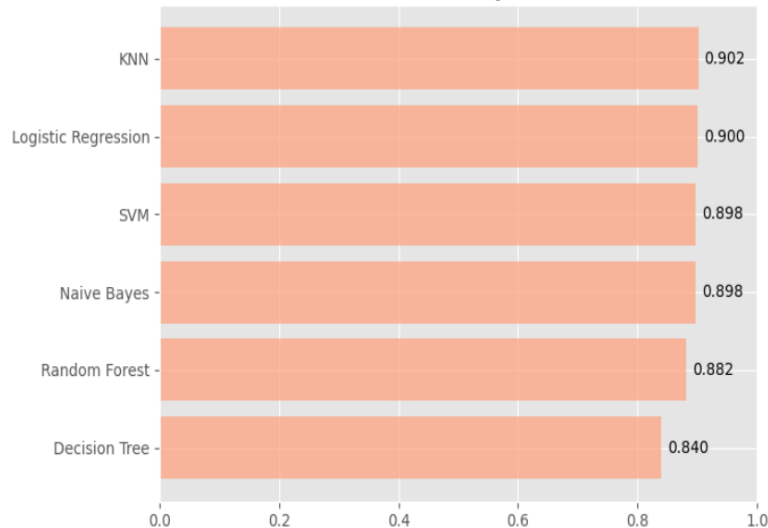
In all models we used a train-test split to allocate the data into a testing and a training set

Model Results and Assessment: A process of three sub-steps

Random Forest best model after tuning, with chest pain the most critical predictive feature

Step 4a: Initial assessment

F1-Score Comparison



- KNN model seems to perform better overall
- It is followed by Logistic Regression and SVM with minor differences
- Decision tree with lower classification performance

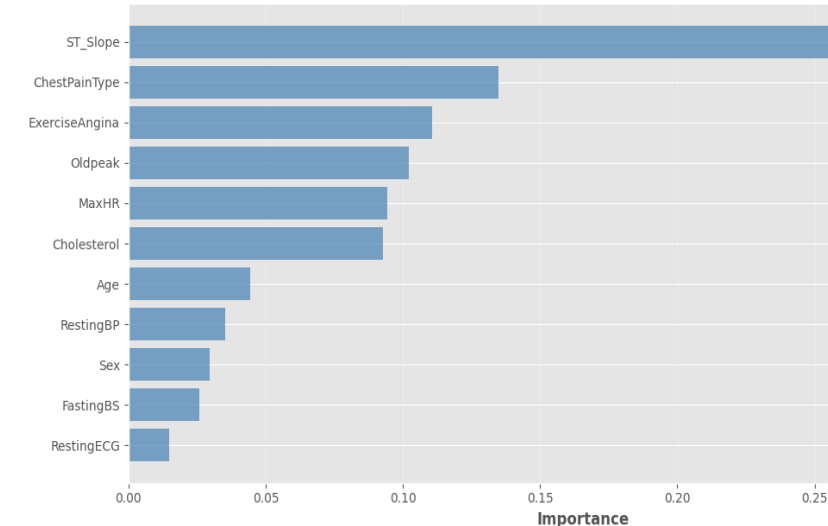
Step 4b: Fine-tuning

Model Name	F1 Score
Random Forest (RF)	0.8836
Support Vector Machine (SVM)	0.8834
K Nearest Neighbors (KNN)	0.8812

- Model adjustments through five-fold cross-validation and hyperparameter tuning
- Following tuning, Random Forest and SVM appear to be the best models classifying people best equally well

Step 4c: Feature significance

Feature Importance - Random Forest



- Type of chest pain and exercise angina the features with highest predictive importance
- Demographics such as age and gender seem to have smaller predictive significance
- Unexpectedly, total cholesterol seems to also have rather average importance

As a Summary

In this project we have :

- Tried to identify ways to classify a population into those that will develop heart disease and those not
- Deep-dived into different types of Machine Learning models to enable us achieve our goal
- Leveraged Python and its vast capabilities to reach robust results and test their accuracy

A journey through Machine Learning and Python to familiarize with Big Data in Decision-Making